1. General Description

The EM74LVC1T45 and EM74LVCH1T45 are single bit, dual supply transceivers with 3-state outputs that enable bidirectional level translation. They feature two 1-bit input-output ports (A and B), a direction control input (DIR) and dual supply pins ($V_{CC(A)}$ and $V_{CC(B)}$). Both $V_{CC(A)}$ and $V_{CC(B)}$ can be supplied at any voltage between 1.2 V and 5.5 V making the device suitable for translating between any of the low voltage nodes (1.2 V, 1.5 V, 1.8 V, 2.5 V, 3.3 V and 5.0 V). Pins A and DIR are referenced to $V_{CC(A)}$ and pin B is referenced to $V_{CC(B)}$. A HIGH on DIR allows transmission from A to B and a LOW on DIR allows transmission from B to A.

The devices are fully specified for partial power-down applications using IOFF. The IOFF circuitry disables the output, preventing any damaging backflow current through the device when it is powered down. In suspend mode when either $V_{CC(A)}$ or $V_{CC(B)}$ are at GND level, both A port and B port are in the high-impedance OFF-state.

Active bus hold circuitry in the EM74LVCH1T45 holds unused or floating data inputs at a valid logic level.

2. Features and Benefits

- Wide supply voltage range:
  - $V_{CC(A)}$: 1.2 V to 5.5 V
  - $V_{CC(B)}$: 1.2 V to 5.5 V
- High noise immunity
- Suspend mode
- Latch-up performance exceeds 100 mA
- Inputs accept voltages up to 5.5 V
- Low power consumption: 16 µA maximum I_{CC}
- I_{OFF} circuitry provides partial Power-down mode operation
- Complies with JEDEC standards:
  - JESD8-7 (1.2 V to 1.95 V)
  - JESD8-5 (1.8 V to 2.7 V)
  - JESD8C (2.7 V to 3.6 V)
  - JESD36 (4.5 V to 5.5 V)
- ESD protection:
  - HBM ANSI/ESDA/JEDEC JS-001 Class 3A exceeds 2000 V
  - CDM ANSI/ESDA/JEDEC JS-002 Class C3 exceeds 1000 V
- Multiple package options

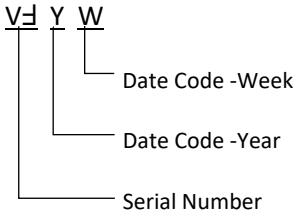
3. Ordering Information

Table 1. Ordering information

Type number	Topside marking	Package		
		Name	Description	Quantity
EM74LVC1T45GV	V∃YW	SOT23-6L	SOT23 package, 6 pins 2.92 mm × 1.6 mm; 1.25 mm (Max) height	3000
EM74LVC1T45GW	V∃YW	SOT363	SOT363 package, 6 pins 2.1 mm × 1.25 mm; 1.1 mm (Max) height	3000
EM74LVC1T45GS	V∃	DFN1x1-6L	DFN1×1 package, 6 pins 1 mm × 1 mm; 0.42 mm (Max) height	3000
EM74LVC1T45GM	V∃YW	DFN1x1.45-6L	DFN1.45×1 package, 6 pins 1.45 mm × 1 mm; 0.6 mm (Max) height	3000
EM74LVC1T45YZP	V∃	BGA1.39x0.89-6B	BGA 1.39 × 0.89 package， 6 bumps 1.39mm × 0..89mm; 0.5 mm (Max) height	3000

MARKING INFORMATION

NOTE: YW = Date Code.



4. Function Diagram

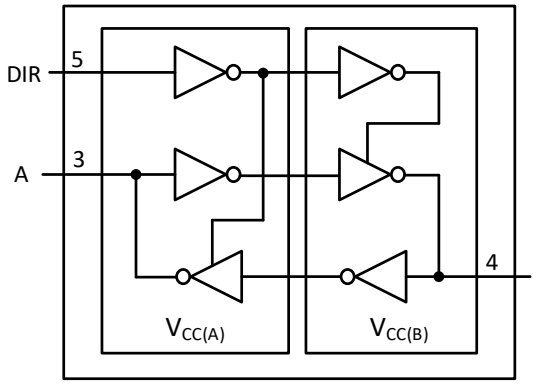


Fig. 1. Logic symbol

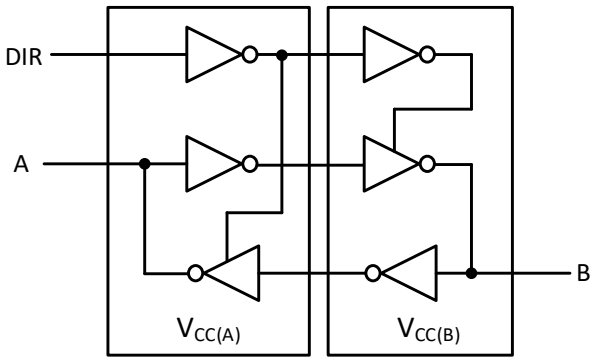


Fig. 2. Logic diagram

5. Pinning Information

5.1. Pinning

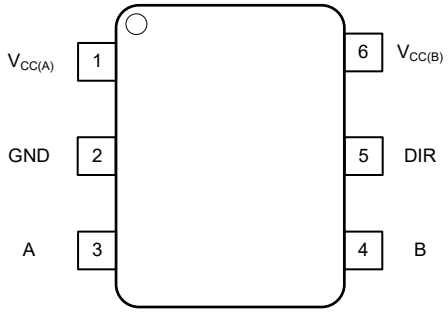


Fig. 3. Top view pin configuration SOT23-6L and SOT363

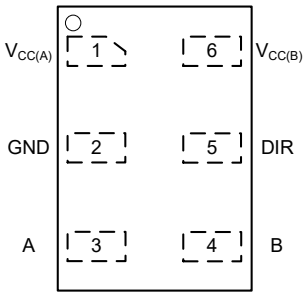


Fig. 4. Top view pin configuration DFN6L

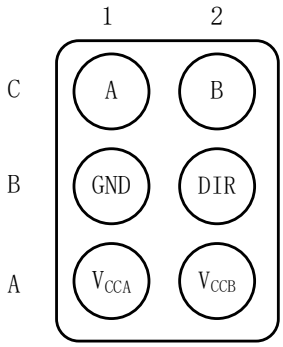


Fig. 5. Bottom view pin configuration BGA-6B

5.2. Pin description

Table 2. Pin description

Symbol	Pin		Description
	SOT23-6L,SOT363,DFN6L	BGA-6B	
$V_{CC(A)}$	1	A1	supply voltage port A and DIR
GND	2	B1	Ground (0V)
A	3	C1	Data input or output
B	4	C2	Data input or output
DIR	5	B2	Direction control
$V_{CC(B)}$	6	A2	Supply voltage port B

[1] All GND pins must be connected to ground (0V).

6. Functional Description

Table 3. Function table

Supply voltage	Input	Input/Output [1]	
$V_{CC(A)}$, $V_{CC(A)}$	DIR	A	B
1.2 V to 5.5 V	L	A=B	Input
1.2 V to 5.5 V	H	input	B=A
GND [2]	X	Z	Z

[1] If at least one of $V_{CC(A)}$ or $V_{CC(B)}$ is at GND level, the device goes into suspend mode.
[2] The An inputs/outputs, DIR and OE input circuit is referenced to $V_{CC(A)}$; The Bn inputs/outputs circuit is referenced to $V_{CC(B)}$.

7. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

Table 4. Absolute Maximum Ratings

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

Symbol	Parameter	Conditions	Min	Max	Unit
$V_{CC(A)}$	supply voltage A		-0.5	6.5	V
$V_{CC(B)}$	supply voltage B		-0.5	6.5	V
I_{IK}	input clamping current	$V_I < 0$ V	-20		mA
V_I	input voltage	[1]	-0.5	6.5	V
I_{OK}	output clamping current	$V_O < 0$ V	-50		mA
V_O	output voltage	Activemode [1][2][3]	-0.5	V_{CCO} + 0.5	V
		Suspend or 3-state mode [1]	-0.5	4.6	V
I_O	output current	V_O = 0 V to V_{CCO}		±50	mA
I_{CC}	supply current	$I_{CC(A)}$ or $I_{CC(B)}$		100	mA
I_{GND}	ground current		-100		mA
P_{tot}	total power dissipation	T_{amb} = -40 °C to + 125 °C		250	mW
T_{stg}	storage temperature		-65	150	°C

[1] The input and output voltage ratings may be exceeded if the input and output current ratings are observed.
[2] V_{CCO} is the supply voltage associated with the output port.
[3] V_{CCO} + 0.5 V should not exceed 6.5 V.

8. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

Table 5. Recommended Operating Conditions

Symbol	Parameter	Conditions	Min	Max	Unit
$V_{CC(A)}$	supply voltage A		1.2	5.5	V
$V_{CC(B)}$	supply voltage B		1.2	5.5	V
V_I	input voltage		0	5.5	V
V_O	output voltage	Active mode [1]	0	V_{CCO}	V
		Suspend or 3-state mode	0	5.5	V
T_{amb}	ambient temperature		-40	125	°C
Δt/ΔV	input transition rise and fall rate	V_{CCI} = 1.2 V [2]		20	ns/V
		V_{CCI} = 1.4 V to 1.95 V		20	ns/V
		V_{CCI} = 2.3 V to 2.7 V		20	ns/V
		V_{CCI} = 3 V to 3.6 V		10	ns/V
		V_{CC} = 4.5 V to 5.5 V		5	ns/V

[1] V_{CCO} is the supply voltage associated with the output port.
[2] V_{CCI} is the supply voltage associated with the input port.

9. Static Characteristics

Table 6. Static characteristics

At recommended operating conditions. Voltages are referenced to GND (ground = 0 V). Typical values measured at T_{amb} = 25°C (unless otherwise noted).

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ	Max	Min	Max	
V_{IH}	HIGH-level input voltage	data input [1]						
		V_{CCI} = 1.2 V	$0.8V_{CCI}$			$0.8V_{CCI}$		V
		V_{CCI} = 1.4 V to 1.95 V	$0.65V_{CCI}$			$0.65V_{CCI}$		V
		V_{CCI} = 2.3 V to 2.7 V	1.7			1.7		V
		V_{CCI} = 3 V to 3.6 V	2.2			2.2		V
		V_{CCI} = 4.5 V to 5.5 V	$0.7V_{CCI}$			$0.7V_{CCI}$		V
		DIR input						
		V_{CCI} = 1.2 V	$0.8V_{CC(A)}$			$0.8V_{CC(A)}$		V
		V_{CCI} = 1.4 V to 1.95 V	$0.65V_{CC(A)}$			$0.65V_{CC(A)}$		V
		V_{CCI} = 2.3 V to 2.7 V	1.7			1.7		V
		V_{CCI} = 3 V to 3.6 V	2.0			2.0		V
		V_{CCI} = 4.5 V to 5.5 V	$0.7V_{CC(A)}$			$0.7V_{CC(A)}$		V
V_{IL}	LOW-level input voltage	data input [1]						
		V_{CCI} = 1.2 V			$0.2V_{CCI}$		$0.2V_{CCI}$	V
		V_{CCI} = 1.4 V to 1.95 V			0.35 V_{CCI}		0.35 V_{CCI}	V
		V_{CCI} = 2.3 V to 2.7 V			0.7		0.7	V
		V_{CCI} = 3 V to 3.6 V			0.8		0.8	V
		V_{CCI} = 4.5 V to 5.5 V			$0.3V_{CCI}$		$0.3V_{CCI}$	V
		DIR, input						
		V_{CCI} = 1.2 V			$0.2V_{CC(A)}$		$0.2V_{CC(A)}$	V
		V_{CCI} = 1.4 V to 1.95 V			0.35 $V_{CC(A)}$		0.35 $V_{CC(A)}$	V
		V_{CCI} = 2.3 V to 2.7 V			0.7		0.7	V
		V_{CCI} = 3 V to 3.6 V			0.8		0.8	V
		V_{CCI} = 4.5 V to 5.5 V			$0.3V_{CC(A)}$		$0.3V_{CC(A)}$	V

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ	Max	Min	Max	
V_{OH}	HIGH-level output voltage	$V_I = V_{IH}$ [2]						
		I_O = -100 µA; V_{CCO} = 1.2 V to 4.5 V	V_{CCO} - 0.1			V_{CCO} - 0.1		V
		I_O = -3 mA; V_{CCO} = 1.4 V	1.0			1.0		V
		I_O = -5.5 mA; V_{CCO} = 1.65 V	1.2			1.2		V
		I_O = -10 mA; V_{CCO} = 2.3 V	1.9			1.8		V
		I_O = -20 mA; V_{CCO} = 3.0 V	2.4			2.3		V
		I_O = -32 mA; V_{CCO} = 4.5 V	3.8			3.7		V
V_{OL}	LOW-level output voltage	$V_I = V_{IL}$ [2]						
		I_O = 100 µA; V_{CCO} = 1.2 V to 4.5 V			0.1		0.1	V
		I_O = 3 mA; V_{CCO} = 1.4 V			0.3		0.3	V
		I_O = 5.5 mA; V_{CCO} = 1.65 V			0.45		0.45	V
		I_O = 10 mA; V_{CCO} = 2.3 V			0.3		0.4	V
		I_O = 20 mA; V_{CCO} = 3.0 V			0.55		0.55	V
		I_O = 32 mA; V_{CCO} = 4.5 V			0.55		0.6	V
I_I	input leakage current	DIR input; V_I = 0V to 5.5V ; V_{CCI} = 1.2V to 5.5V			±2		±10	µA
I_{OZ}	OFF-state output current	A or B port; V_O = 0 V or V_{CCO}; V_{CCO} = 1.2 V to 5.5 V			±2		±10	µA
I_{OFF}	power-off leakage current	A port; V_I or V_O = 0 V to 5.5 V; $V_{CC(A)}$ = 0 V; $V_{CC(B)}$ = 1.2 V to 5.5 V			±2		±10	µA
		B port; V_I or V_O = 0 V to 5.5 V; $V_{CC(B)}$ = 0 V; $V_{CC(A)}$ = 1.2 V to 5.5 V			±2		±10	µA
I_{BHL}	bus hold LOW current	A or B port [1]						
		V_I = 0.49 V; V_{CCI} = 1.4 V	15			10		µA
		V_I = 0.58 V; V_{CCI} = 1.65 V	25			20		µA
		V_I = 0.70 V; V_{CCI} = 2.3 V	45			45		µA
		V_I = 0.80 V; V_{CCI} = 3.0 V	100			80		µA
		V_I = 1.35 V; V_{CCI} = 4.5 V	100			100		µA
I_{BHH}		A or B port [1]						

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ	Max	Min	Max	
	bus hold HIGH current	V_I = 0.49 V; V_{CCI} = 1.4 V	-15			-10		µA
		V_I = 1.07 V; V_{CCI} = 1.65 V	-25			-20		µA
		V_I = 1.70 V; V_{CCI} = 2.3 V	-45			-45		µA
		V_I = 2.00 V; V_{CCI} = 3.0 V	-100			-80		µA
		V_I = 3.15 V; V_{CCI} = 4.5 V	-100			-100		µA
I_{BHLO}	bus hold LOW overdrive current	A or B port [1][3]						
		V_{CCI} = 1.6 V	125			125		µA
		V_{CCI} = 1.95 V	200			200		µA
		V_{CCI} = 2.7 V	300			300		µA
		V_{CCI} = 3.6 V	500			500		µA
		V_{CCI} = 5.5 V	900			900		µA
I_{BHHO}	bus hold HIGH overdrive current	A or B port [1][3]						
		V_{CCI} = 1.6 V	-125			-125		µA
		V_{CCI} = 1.95 V	-200			-200		µA
		V_{CCI} = 2.7 V	-300			-300		µA
		V_{CCI} = 3.6 V	-500			-500		µA
		V_{CCI} = 5.5 V	-900			-900		µA
I_{CC}	supply current	A port; V_I = 0 V or V_{CCI}; I_O = 0 A [1]						
		$V_{CC(A)}$, $V_{CC(B)}$=1.2V to 5.5V			8		8	µA
		$V_{CC(A)}$, $V_{CC(B)}$=1.65V to 5.5V			3		3	µA
		$V_{CC(A)}$ = 5.5V; $V_{CC(B)}$ = 0V			2		2	µA
		$V_{CC(A)}$ = 0V; $V_{CC(B)}$ = 5.5V	-2			-2		µA
		B port; V_I = 0 V or V_{CCI}; I_O = 0 A						
		$V_{CC(A)}$, $V_{CC(B)}$=1.2V to 5.5V			8		8	µA
		$V_{CC(A)}$, $V_{CC(B)}$=1.65V to 5.5V			3		3	
		$V_{CC(B)}$ = 0V; $V_{CC(A)}$ = 5.5V			2		2	µA
		$V_{CC(B)}$ = 5.5V; $V_{CC(A)}$ = 0V	-2			-2		µA
		A plus B port ($I_{CC(A)}$ + $I_{CC(B)}$); I_O = 0 A; V_I = 0 V or V_{CCI}						
		$V_{CC(A)}$, $V_{CC(B)}$=1.2V to 5.5V			16		16	µA
		$V_{CC(A)}$, $V_{CC(B)}$=1.65V to 5.5V			4		4	
ΔI_{CC}	additional	per input; $V_{CC(A)}$, $V_{CC(B)}$ = 3.0V to 5.5V						

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ	Max	Min	Max	
	supply current	DIR input; DIR input at $V_{CC(A)}$ -0.6V; A port at $V_{CC(A)}$ or GND; B port = open			50		75	µA
		A port; A port at $V_{CC(A)}$ - 0.6 V; DIR at $V_{CC(A)}$; B port = open [4]			50		75	µA
		B port; B port at $V_{CC(B)}$ - 0.6 V; DIR at $V_{CC(B)}$; A port = open [4]			50		75	µA
C_I	input capacitance	DIR input; V_I = 0 V or 3.3 V; $V_{CC(A)}$) = $V_{CC(B)}$ = 3.3 V		2				pF
$C_{I/O}$	input/ output capacitance	A and B port; suspend mode; V_O = 3.3 V or 0 V; $V_{CC(A)}$ = $V_{CC(B)}$ = 3.3 V		6				pF

[1] V_{CCI} is the supply voltage associated with the data input port.
[2] V_{CCO} is the supply voltage associated with the output port.
[3] To guarantee the node switches, an external driver must source/sink at least I_{BHLO} / I_{BHHO} when the input is in the range V_{IL} to V_{IH}.
[4] For non bus hold parts only (EM74LVC1T45).

10. Dynamic Characteristics

Table 7. Dynamic characteristics at VCC(A) = 1.2V and T_{amb} = 25 °C

Voltages are referenced to GND (ground = 0 V).For test circuit see Fig. 8. for waveforms see Fig. 6 and Fig. 7.

Symbol	Parameter	Conditions	$V_{CC(B)}$						Unit
			1.2 V	1.5 V	1.8 V	2.5 V	3.3 V	5.0 V	
t_{PLH}	LOW to HIGH propagation time	A to B	10.6	8.1	7.0	5.8	5.3	5.1	ns
		B to A	10.6	9.5	9.0	8.5	8.3	8.2	ns
t_{PHL}	HIGH to LOW propagation time	A to B	10.1	7.1	6.0	5.3	5.2	5.4	ns
		B to A	10.1	8.6	8.1	7.8	7.6	7.6	ns
t_{PHZ}	HIGH to OFF-state propagation delay	DIR to A	9.4	9.4	9.4	9.4	9.4	9.4	ns
		DIR to B	12.0	9.4	9.0	7.8	8.4	7.9	ns
t_{PLZ}	LOW to OFF-state propagation delay	DIR to A	7.1	7.1	7.1	7.1	7.1	7.1	ns
		DIR to B	9.5	7.8	7.7	6.9	7.6	7.0	ns
t_{PZH}	OFF-state to HIGH propagation delay	DIR to A[1]	20.1	17.3	16.7	15.4	15.9	15.2	ns
		DIR to B[1]	17.7	15.2	14.1	12.9	12.4	12.2	ns
t_{PZL}	OFF-state to LOW propagation delay	DIR to A[1]	22.1	18.0	17.1	15.6	16.0	15.5	ns
		DIR to B[1]	19.5	16.5	15.4	14.7	14.6	14.8	ns

[1] Calculate the enable times for the EM74LVC1T45; EM74LVCH1T45 using the following formulas:

- t_{PZH} (DIR to A) = t_{PLZ} (DIR to B) + t_{PLH} (B to A); t_{PZL} (DIR to A) = t_{PHZ} (DIR to B) + t_{PHL} (B to A)
- t_{PZH} (DIR to B) = t_{PLZ} (DIR to A) + t_{PLH} (A to B); t_{PZL} (DIR to B) = t_{PHZ} (DIR to A) + t_{PHL} (A to B)

In a bidirectional application, these enable times provide the maximum delay from the time the DIR bit is switched until an output is expected. For example, if the EM74LVC1T45; EM74LVCH1T45 initially is transmitting from A to B, then the DIR bit is switched, the B port of the device must be disabled before presenting it with an input. After the B port has been disabled, an input signal applied to it appears on the corresponding A port after the specified propagation delay.

Table 8. Typical power dissipation capacitance at $V_{CC(A)}$ = $V_{CC(B)}$ and T_{amb} = 25 °C

Voltages are referenced to GND (ground = 0 V). [1][2]

Symbol	Parameter	Conditions	$V_{CC(A)}$ and $V_{CC(B)}$				Unit
			1.8 V	2.5 V	3.3 V	5.5 V	
C_{PD}	power dissipation capacitance	A port: (direction A to B); B port: (direction B to A)	2	3	3	4	pF
		A port: (direction B to A); B port: (direction A to B)	15	16	16	18	pF

[1] C_{PD} is used to determine the dynamic power dissipation (P_D in µW).

$P_D = C_{PD} \times V_{CC}^2 \times f_i \times N + \Sigma(C_L \times V_{CC}^2 \times f_o)$ where:

f_i = input frequency in MHz;

f_o = output frequency in MHz;

C_L = load capacitance in pF;

V_{CC} = supply voltage in V;

N = number of inputs switching;

$\Sigma(C_L \times V_{CC}^2 \times f_o)$ = sum of the outputs.

[2] f_i = 10 MHz; V_I = GND to V_{CC}; $t_r = t_f$ = 1 ns; C_L = 0 pF; R_L = ∞ Ω.

Table 9. Dynamic characteristics for temperature range -40 °C to +85 °C

Voltages are referenced to GND (ground = 0 V).For test circuit see Fig. 8. for waveforms see Fig. 6 and Fig. 7.

Symbol	Parameter	Conditions	$V_{CC(B)}$										Unit
			1.5V±0.1V		1.8V±0.15V		2.5V±0.2V		3.3V±0.3V		5.0V±0.5V		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
$V_{CC(A)}$ = 1.5 V ± 0.1 V													
t_{PLH}	LOW to HIGH propagation time	A to B	2.8	21.3	2.4	17.6	2.0	13.5	1.7	11.8	1.6	10.5	ns
		B to A	2.8	21.3	2.6	19.1	2.3	14.9	2.3	12.4	2.2	12.0	ns
t_{PHL}	HIGH to LOW propagation delay	A to B	2.6	19.3	2.2	15.3	1.8	11.8	1.7	10.9	1.7	10.8	ns
		B to A	2.6	19.3	2.4	17.3	2.3	13.2	2.2	11.3	2.3	11.0	ns
t_{PHZ}	HIGH to OFF-state propagation delay	DIR to A	3.0	18.7	3.0	18.7	3.0	18.7	3.0	18.7	3.0	18.7	ns
		DIR to B	3.5	24.8	3.5	23.6	3.0	11.0	3.3	11.3	2.8	10.3	ns
t_{PLZ}	LOW to OFF-state propagation delay	DIR to A	2.4	11.4	2.4	11.4	2.4	11.4	2.4	11.4	2.4	11.4	ns
		DIR to B	2.8	18.3	3.0	17.2	2.5	9.4	3.0	10.1	2.5	9.4	ns
t_{PZH}	OFF-state to HIGH propagation delay	DIR to A [1]		39.6		36.3		24.3		22.5		21.4	ns
		DIR to B [1]		32.7		29.0		24.9		23.2		21.9	ns
t_{PZL}	OFF-state to LOW propagation delay	DIR to A [1]		44.1		40.9		24.2		22.6		21.3	ns
		DIR to B [1]		38.0		34.0		30.5		29.6		29.5	ns
$V_{CC(A)}$ = 1.8 V ± 0.15 V													
t_{PLH}	LOW to HIGH propagation delay	A to B	2.6	19.1	2.2	17.7	2.2	9.3	1.7	7.2	1.4	6.8	ns
		B to A	2.4	17.6	2.2	17.7	2.3	16.0	2.1	15.5	1.9	15.1	ns
t_{PHL}	HIGH to LOW propagation delay	A to B	2.4	17.3	2.0	14.3	1.6	8.5	1.8	7.1	1.7	7.0	ns
		B to A	2.2	15.3	2.0	14.3	2.1	12.9	2.0	12.6	1.8	12.2	ns
t_{PHZ}	HIGH to OFF-state propagation delay	DIR to A	2.9	17.1	2.9	17.1	2.9	17.1	2.9	17.1	2.9	17.1	ns
		DIR to B	3.2	24.1	3.2	21.9	2.7	11.5	3.0	10.3	2.5	8.2	ns

t_{PLZ}	LOW to OFF-state propagation delay	DIR to A	2.4	10.5	2.4	10.5	2.4	10.5	2.4	10.5	2.4	10.5	ns
		DIR to B	2.5	17.6	2.6	16.0	2.2	9.2	2.7	8.4	2.4	6.4	ns
t_{PZH}	OFF-state to HIGH propagation delay	DIR to A [1]		35.2		33.7		25.2		23.9		21.8	ns
		DIR to B [1]		29.6		28.2		19.8		17.7		17.3	ns
t_{PZL}	OFF-state to LOW propagation delay	DIR to A [1]		39.4		36.2		24.4		22.9		20.4	ns
		DIR to B [1]		34.4		31.4		25.6		24.2		24.1	ns
$V_{CC(A)}$ = 2.5 V ± 0.2 V													
t_{PLH}	LOW to HIGH propagation delay	A to B	2.3	17.9	2.3	16.0	1.5	8.5	1.3	6.2	1.1	4.8	ns
		B to A	2.0	13.5	2.2	9.3	1.5	8.5	1.4	8.0	1.0	7.5	ns
t_{PHL}	HIGH to LOW propagation delay	A to B	2.3	15.8	2.1	12.9	1.4	7.5	1.3	5.4	0.9	4.6	ns
		B to A	1.8	11.8	1.9	8.5	1.4	7.5	1.3	7.0	0.9	6.2	ns
t_{PHZ}	HIGH to OFF-state propagation delay	DIR to A	2.1	8.1	2.1	8.1	2.1	8.1	2.1	8.1	2.1	8.1	ns
		DIR to B	3.0	22.5	3.0	21.4	2.5	11.0	2.8	9.3	2.3	6.9	ns
t_{PLZ}	LOW to OFF-state propagation delay	DIR to A	1.7	5.8	1.7	5.8	1.7	5.8	1.7	5.8	1.7	5.8	ns
		DIR to B	2.3	14.6	2.5	13.2	2.0	9.0	2.5	8.4	1.8	5.3	ns
t_{PZH}	OFF-state to HIGH propagation delay	DIR to A [1]		28.1		22.5		17.5		16.4		12.8	ns
		DIR to B [1]		23.7		21.8		14.3		12.0		10.6	ns
t_{PZL}	OFF-state to LOW propagation delay	DIR to A [1]		34.3		29.9		18.5		16.3		13.1	ns
		DIR to B [1]		23.9		21.0		15.6		13.5		12.7	ns
$V_{CC(A)}$ = 3.3 V ± 0.3 V													
t_{PLH}	LOW to HIGH propagation delay	A to B	2.3	17.1	2.1	15.5	1.4	8.0	0.8	5.6	0.7	4.4	ns
		B to A	1.7	11.8	1.7	7.2	1.3	6.2	0.7	5.6	0.6	5.4	ns
t_{PHL}	HIGH to LOW propagation delay	A to B	2.2	15.6	2.0	12.6	1.3	7.0	0.8	5.0	0.7	4.0	ns
		B to A	1.7	10.9	1.8	7.1	1.3	5.4	0.8	5.0	0.7	4.5	ns

t_{PHZ}	HIGH to OFF-state propagation delay	DIR to A	2.3	7.3	2.3	7.3	2.3	7.3	2.3	7.3	2.7	7.3	ns
		DIR to B	2.9	18.0	2.9	16.5	2.3	10.1	2.7	8.6	2.2	6.3	ns
t_{PLZ}	LOW to OFF-state propagation delay	DIR to A	2.0	5.6	2.0	5.6	2.0	5.6	2.0	5.6	2.0	5.6	ns
		DIR to B	2.3	13.6	2.4	12.5	1.9	7.8	2.3	7.1	1.7	4.9	ns
t_{PZH}	OFF-state to HIGH propagation delay	DIR to A [1]		25.4		19.7		14.0		12.7		10.3	ns
		DIR to B [1]		22.7		21.1		13.6		11.2		10.0	ns
t_{PZL}	OFF-state to LOW propagation delay	DIR to A [1]		28.9		23.6		15.5		13.6		10.8	ns
		DIR to B [1]		22.9		19.9		14.3		12.3		11.3	ns
$V_{CC(A)}$ = 5.0 V ± 0.5 V													
t_{PLH}	LOW to HIGH propagation delay	A to B	2.2	16.6	1.9	15.1	1.0	7.5	0.7	5.4	0.5	3.9	ns
		B to A	1.6	10.5	1.4	6.8	1.0	4.8	0.7	4.4	0.5	3.9	ns
t_{PHL}	HIGH to LOW propagation delay	A to B	2.3	15.3	1.8	12.2	1.0	6.2	0.7	4.5	0.5	3.5	ns
		B to A	1.7	10.8	1.7	7.0	0.9	4.6	0.7	4.0	0.5	3.5	ns
t_{PHZ}	HIGH to OFF-state propagation delay	DIR to A	1.7	5.4	1.7	5.4	1.7	5.4	1.7	5.4	1.7	5.4	ns
		DIR to B	2.9	17.3	2.9	16.1	2.3	9.7	2.7	8.0	2.5	5.7	ns
t_{PLZ}	LOW to OFF-state propagation delay	DIR to A	1.4	3.7	1.4	3.7	1.3	3.7	1.0	3.7	0.9	3.7	ns
		DIR to B	2.3	13.1	2.4	12.1	1.9	7.4	2.3	7.0	1.8	4.5	ns
t_{PZH}	OFF-state to HIGH propagation delay	DIR to A [1]		23.6		18.9		12.2		11.4		8.4	ns
		DIR to B [1]		20.3		18.8		11.2		9.1		7.6	ns
t_{PZL}	OFF-state to LOW propagation delay	DIR to A [1]		28.1		23.1		14.3		12.0		9.2	ns
		DIR to B [1]		20.7		17.6		11.6		9.9		8.9	ns

[1] Calculate the enable times for the EM74LVC1T45; EM74LVCH1T45 using the following formulas:

- t_{PZH} (DIR to A) = t_{PLZ} (DIR to B) + t_{PLH} (B to A); t_{PZL} (DIR to A) = t_{PHZ} (DIR to B) + t_{PHL} (B to A)
- t_{PZH} (DIR to B) = t_{PLZ} (DIR to A) + t_{PLH} (A to B); t_{PZL} (DIR to B) = t_{PHZ} (DIR to A) + t_{PHL} (A to B)

In a bidirectional application, these enable times provide the maximum delay from the time the DIR bit is switched until an output is expected. For example, if the EM74LVC1T45; EM74LVCH1T45 initially is transmitting from A to B, then the DIR bit is switched, the B port of the device must be disabled before presenting it with an input. After the B port has been disabled, an input signal applied to it appears on the corresponding A port after the specified propagation delay.

Table 9. Dynamic characteristics for temperature range -40 °C to +125 °C.

Voltages are referenced to GND (ground = 0 V).For test circuit see Fig. 8. for waveforms see Fig. 6 and Fig. 7.

Symbol	Parameter	Conditions	$V_{CC(B)}$										Unit
			1.5V±0.1V		1.8V±0.15V		2.5V±0.2V		3.3V±0.3V		5.0V±0.5V		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
$V_{CC(A)}$ = 1.5 V ± 0.1 V													
t_{PLH}	LOW to HIGH propagation time	A to B	2.5	23.5	2.1	19.4	1.8	14.9	1.5	13.0	1.4	11.6	ns
		B to A	2.5	23.5	2.3	21.1	2.0	16.4	2.0	13.7	1.9	13.2	ns
t_{PHL}	HIGH to LOW propagation delay	A to B	2.3	21.3	1.9	16.9	1.6	13.0	1.5	12.0	1.5	11.9	ns
		B to A	2.3	21.3	2.1	19.1	2.0	14.6	1.9	12.5	2.0	12.1	ns
t_{PHZ}	HIGH to OFF-state propagation delay	DIR to A	2.7	20.6	2.7	20.6	2.7	20.6	2.7	20.6	2.7	20.6	ns
		DIR to B	3.1	27.3	3.1	26.0	2.7	12.1	2.9	12.5	2.5	11.4	ns
t_{PLZ}	LOW to OFF-state propagation delay	DIR to A	2.1	12.6	2.1	12.6	2.1	12.6	2.1	12.6	2.1	12.6	ns
		DIR to B	2.5	20.2	2.7	19.0	2.2	10.4	2.7	11.2	2.2	10.4	ns
t_{PZH}	OFF-state to HIGH propagation delay	DIR to A [1]		43.7		40.1		26.8		24.9		23.6	ns
		DIR to B [1]		36.1		32.0		27.5		25.6		24.2	ns
t_{PZL}	OFF-state to LOW propagation delay	DIR to A [1]		48.6		45.1		26.7		25.0		23.5	ns
		DIR to B [1]		41.9		37.5		33.6		32.6		32.5	ns
$V_{CC(A)}$ = 1.8 V ± 0.15 V													
t_{PLH}	LOW to HIGH propagation delay	A to B	2.3	21.1	1.9	19.5	1.9	10.3	1.5	8.0	1.2	7.5	ns
		B to A	2.1	19.4	1.9	19.5	2.0	17.6	1.8	17.1	1.7	16.7	ns
t_{PHL}	HIGH to LOW propagation delay	A to B	2.1	19.1	1.8	15.8	1.4	9.4	1.6	7.9	1.5	7.7	ns
		B to A	1.9	16.9	1.8	15.8	1.8	14.2	1.8	13.9	1.6	13.5	ns
t_{PHZ}	HIGH to OFF-state propagation delay	DIR to A	2.6	18.9	2.6	18.9	2.6	18.9	2.6	18.9	2.6	18.9	ns
		DIR to B	2.8	26.6	2.8	24.1	2.4	12.7	2.7	11.4	2.2	9.1	ns

t_{PLZ}	LOW to OFF-state propagation delay	DIR to A	2.1	11.6	2.1	11.6	2.1	11.6	2.1	11.6	2.1	11.6	ns
		DIR to B	2.2	19.4	2.3	17.6	1.9	10.2	2.4	9.3	2.1	7.4	ns
t_{PZH}	OFF-state to HIGH propagation delay	DIR to A [1]		38.8		37.1		27.8		26.4		24.1	ns
		DIR to B [1]		32.7		31.1		21.9		19.6		19.1	ns
t_{PZL}	OFF-state to LOW propagation delay	DIR to A [1]		43.5		39.9		26.9		25.3		22.6	ns
		DIR to B [1]		38.0		34.7		28.3		26.8		26.6	ns
$V_{CC(A)}$ = 2.5 V ± 0.2 V													
t_{PLH}	LOW to HIGH propagation delay	A to B	2.0	19.7	2.0	17.6	1.3	9.4	1.1	6.9	0.9	5.3	ns
		B to A	1.8	14.9	1.9	10.3	1.3	9.4	1.2	8.8	0.9	8.3	ns
t_{PHL}	HIGH to LOW propagation delay	A to B	2.0	17.4	1.8	14.2	1.2	8.3	1.1	6.0	0.8	5.1	ns
		B to A	1.6	13.0	1.7	9.4	1.2	8.3	1.1	7.7	0.8	6.9	ns
t_{PHZ}	HIGH to OFF-state propagation delay	DIR to A	1.8	9.0	1.8	9.0	1.8	9.0	1.8	9.0	1.8	9.0	ns
		DIR to B	2.7	24.8	2.7	23.6	2.2	12.1	2.5	10.3	2.0	7.6	ns
t_{PLZ}	LOW to OFF-state propagation delay	DIR to A	1.5	6.4	1.5	6.4	1.5	6.4	1.5	6.4	1.5	6.4	ns
		DIR to B	2.0	16.1	2.2	14.6	1.8	9.9	2.2	9.3	1.6	5.9	ns
t_{PZH}	OFF-state to HIGH propagation delay	DIR to A [1]		31.0		24.9		19.3		18.1		14.2	ns
		DIR to B [1]		26.1		24.0		15.8		13.3		11.7	ns
t_{PZL}	OFF-state to LOW propagation delay	DIR to A [1]		37.8		33.0		20.4		18.0		14.5	ns
		DIR to B [1]		26.4		23.2		17.3		15.0		14.1	ns
$V_{CC(A)}$ = 3.3 V ± 0.3 V													
t_{PLH}	LOW to HIGH propagation delay	A to B	2.0	18.9	1.8	17.1	1.2	8.8	0.7	6.2	0.6	4.9	ns
		B to A	1.5	13.0	1.5	8.0	1.1	6.9	0.6	6.2	0.5	6.0	ns
t_{PHL}	HIGH to LOW propagation delay	A to B	1.9	17.2	1.8	13.9	1.1	7.7	0.7	5.5	0.6	4.4	ns
		B to A	1.5	12.0	1.6	7.9	1.1	6.0	0.7	5.5	0.6	5.0	ns

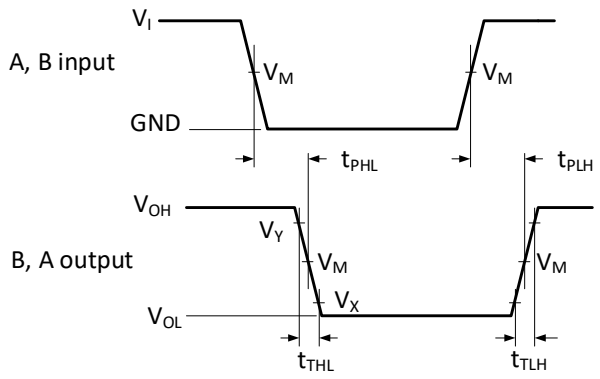
t_{PHZ}	HIGH to OFF-state propagation delay	DIR to A	2.0	8.1	2.0	8.1	2.0	8.1	2.0	8.1	2.4	8.1	ns
		DIR to B	2.6	19.8	2.6	18.2	2.0	11.2	2.4	9.5	1.9	7.0	ns
t_{PLZ}	LOW to OFF-state propagation delay	DIR to A	1.8	6.2	1.8	6.2	1.8	6.2	1.8	6.2	1.8	6.2	ns
		DIR to B	2.0	15.0	2.1	13.8	1.7	8.6	2.0	7.9	1.5	5.4	ns
t_{PZH}	OFF-state to HIGH propagation delay	DIR to A [1]		28.0		21.8		15.5		14.1		11.4	ns
		DIR to B [1]		25.1		23.3		15.0		12.4		11.1	ns
t_{PZL}	OFF-state to LOW propagation delay	DIR to A [1]		31.8		26.1		17.2		15.0		12.0	ns
		DIR to B [1]		25.3		22.0		15.8		13.6		12.5	ns
$V_{CC(A)}$ = 5.0 V ± 0.5 V													
t_{PLH}	LOW to HIGH propagation delay	A to B	1.9	18.3	1.7	16.7	0.9	8.3	0.6	6.0	0.4	4.3	ns
		B to A	1.4	11.6	1.2	7.5	0.9	5.3	0.6	4.9	0.4	4.3	ns
t_{PHL}	HIGH to LOW propagation delay	A to B	2.0	16.9	1.6	13.5	0.9	6.9	0.6	5.0	0.4	3.9	ns
		B to A	1.5	11.9	1.5	7.7	0.8	5.1	0.6	4.4	0.4	3.9	ns
t_{PHZ}	HIGH to OFF-state propagation delay	DIR to A	1.5	6.0	1.5	6.0	1.5	6.0	1.5	6.0	1.5	6.0	ns
		DIR to B	2.6	19.1	2.6	17.8	2.0	10.7	2.4	8.8	2.2	6.3	ns
t_{PLZ}	LOW to OFF-state propagation delay	DIR to A	1.2	4.1	1.2	4.1	1.1	4.1	0.9	4.1	0.8	4.1	ns
		DIR to B	2.0	14.5	2.1	13.4	1.7	8.2	2.0	7.7	1.6	5.0	ns
t_{PZH}	OFF-state to HIGH propagation delay	DIR to A [1]		26.1		20.9		13.5		12.6		9.3	ns
		DIR to B [1]		22.4		20.8		12.4		10.1		8.4	ns
t_{PZL}	OFF-state to LOW propagation delay	DIR to A [1]		31.0		25.5		15.8		13.2		10.2	ns
		DIR to B [1]		22.9		19.5		12.9		11.0		9.9	ns

[1] Calculate the enable times for the EM74LVC1T45; EM74LVCH1T45 using the following formulas:

- t_{PZH} (DIR to A) = t_{PLZ} (DIR to B) + t_{PLH} (B to A); t_{PZL} (DIR to A) = t_{PHZ} (DIR to B) + t_{PHL} (B to A)
- t_{PZH} (DIR to B) = t_{PLZ} (DIR to A) + t_{PLH} (A to B); t_{PZL} (DIR to B) = t_{PHZ} (DIR to A) + t_{PHL} (A to B)

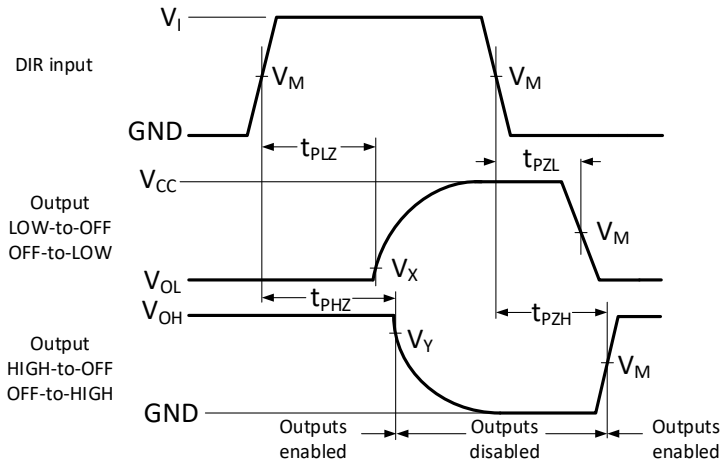
In a bidirectional application, these enable times provide the maximum delay from the time the DIR bit is switched until an output is expected. For example, if the EM74LVC1T45; EM74LVCH1T45 initially is transmitting from A to B, then the DIR bit is switched, the B port of the device must be disabled before presenting it with an input. After the B port has been disabled, an input signal applied to it appears on the corresponding A port after the specified propagation delay.

10.1. Waveforms and test circuit



Measurement points are given in Table 10.
V_{OL} and V_{OH} are typical output voltage levels that occur with the output load.

Fig. 6. The input A, B to output B, A propagation delays



Measurement points are given in Table 10.
V_{OL} and V_{OH} are typical output voltage levels that occur with the output load.

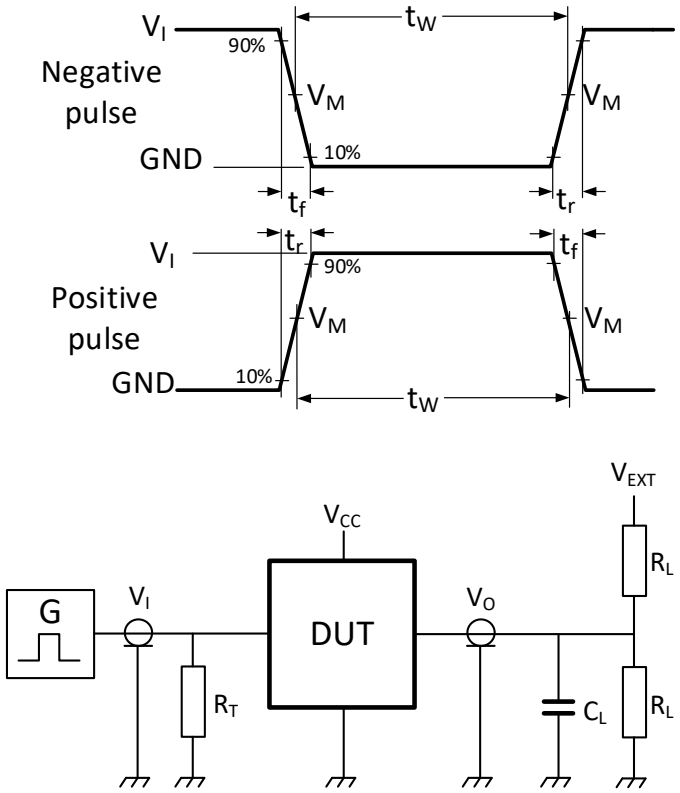
Fig. 7. 3-state enable and disable times

Table 10. Measurement points

Supply voltage	Input [1]	Output [2]		
$V_{CC(A)}$, $V_{CC(B)}$	V_M	V_M	V_X	V_Y
1.2 V to 1.6 V	$0.5V_{CCI}$	$0.5V_{CCO}$	V_{OL} + 0.1 V	V_{OH} - 0.1 V
1.65 V to 2.7 V	$0.5V_{CCI}$	$0.5V_{CCO}$	V_{OL} + 0.15 V	V_{OH} - 0.15 V
3.0 V to 5.5 V	$0.5V_{CCI}$	$0.5V_{CCO}$	V_{OL} + 0.3 V	V_{OH} - 0.3 V

[1] V_{CCI} is the supply voltage associated with the data input port.

[2] V_{CCO} is the supply voltage associated with the output port.



Test data is given in Table 11.
Definitions for test circuit:
R_L = Load resistance.
C_L = Load capacitance including jig and probe capacitance.
R_T = Termination resistance should be equal to the output impedance Z_o of the pulse generator.
V_{EXT} = External voltage for measuring switching times.

Fig. 8. Test circuit for measuring switching times

Table 11. Test data

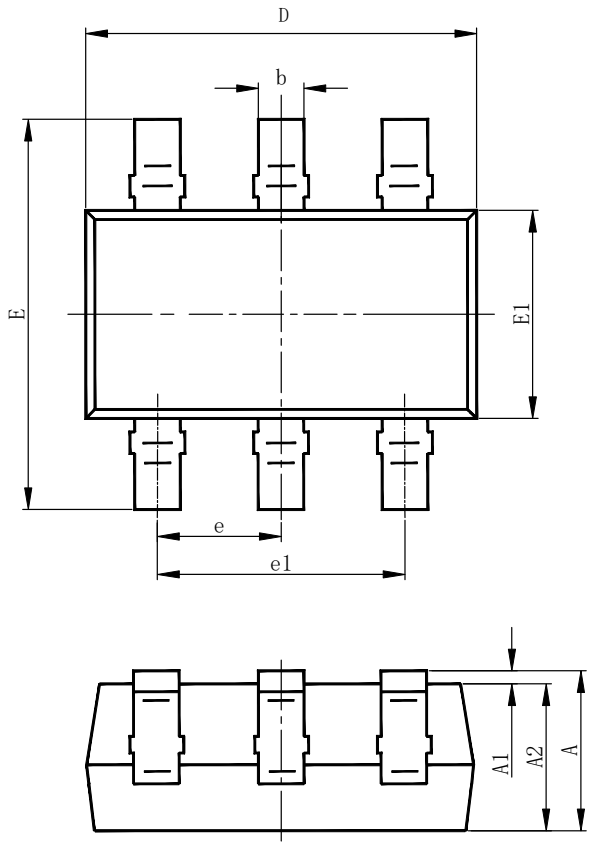
Supply voltage	Input		Load		V_{EXT}		
$V_{CC(A)}$, $V_{CC(B)}$	V_I [1]	$t_r = t_f$	C_L	R_L	t_{PLH}, t_{PHL}	t_{PZH}, t_{PHZ}	t_{PZL}, t_{PLZ} [2]
1.2 V to 5.5 V	V_{CCI}	≤ 2.5 ns	15 pF	2kΩ	open	GND	$2V_{CCO}$

[1] V_{CCI} is the supply voltage associated with the data input port.

[2] V_{CCO} is the supply voltage associated with the output port.

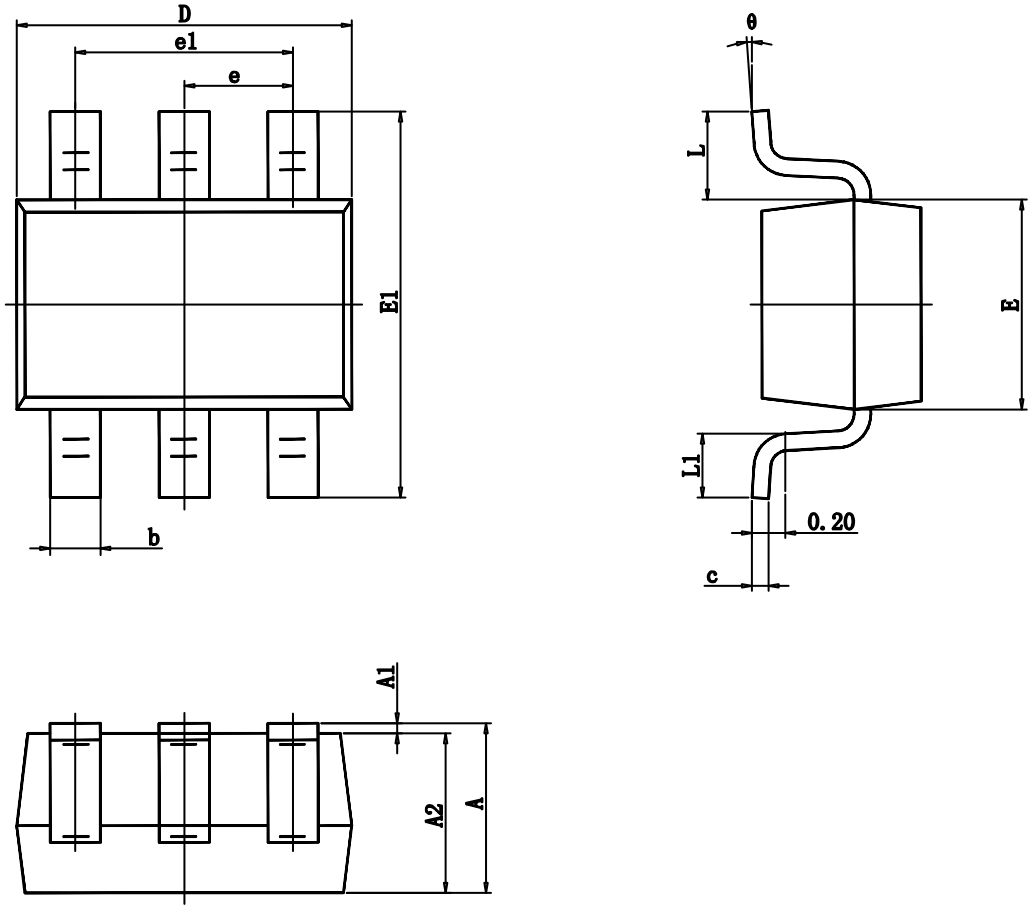
11. Package Outline

SOT23-6L



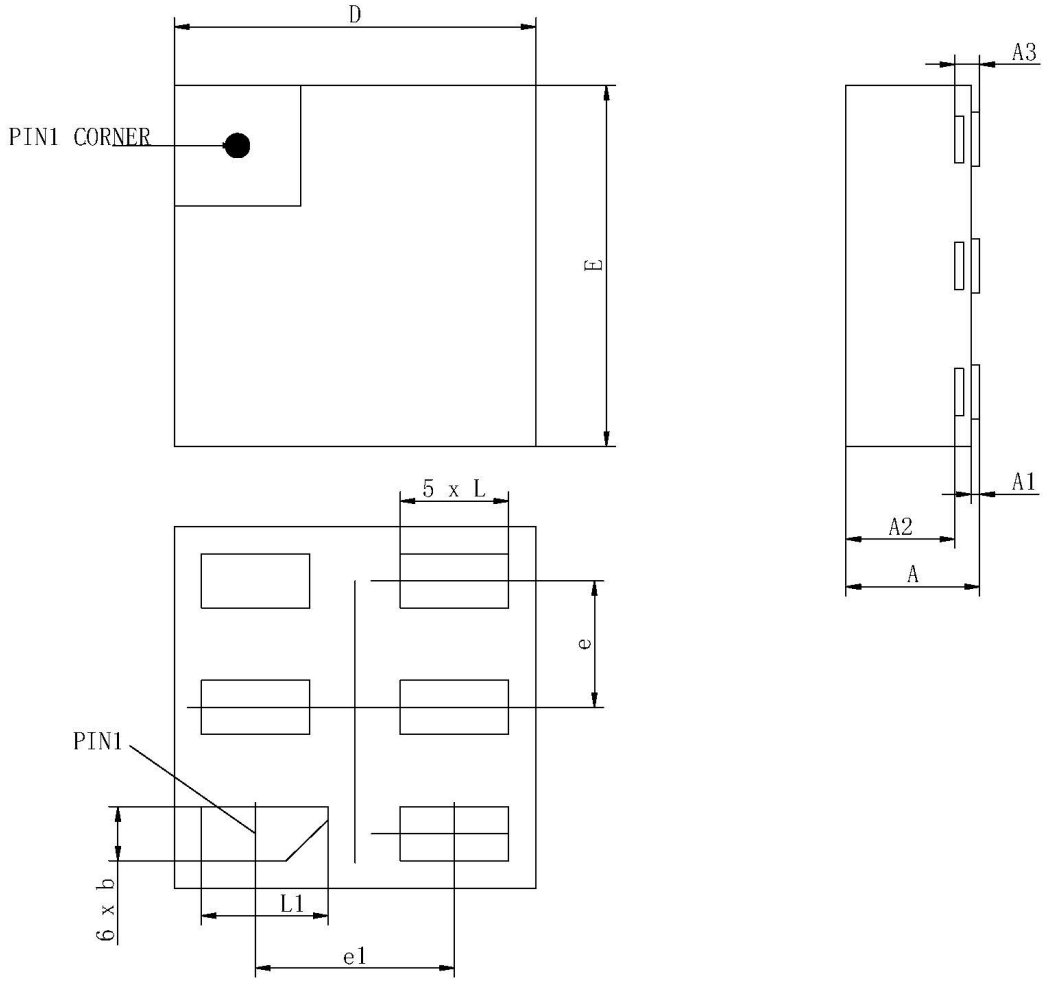
Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	1.050	1.250	0.041	0.049
A1	0.000	0.100	0.000	0.004
A2	1.050	1.150	0.041	0.045
b	0.300	0.500	0.012	0.020
c	0.100	0.200	0.004	0.008
D	2.820	3.020	0.111	0.119
E1	1.500	1.700	0.059	0.067
E	2.650	2.950	0.104	0.116
e	0.950(BSC)		0.037(BSC)	
e1	1.800	2.000	0.071	0.079
L	0.300	0.600	0.012	0.024
L1	0.600REF.		0.024REF.	
θ	0°	8°	0°	8°

SOT363



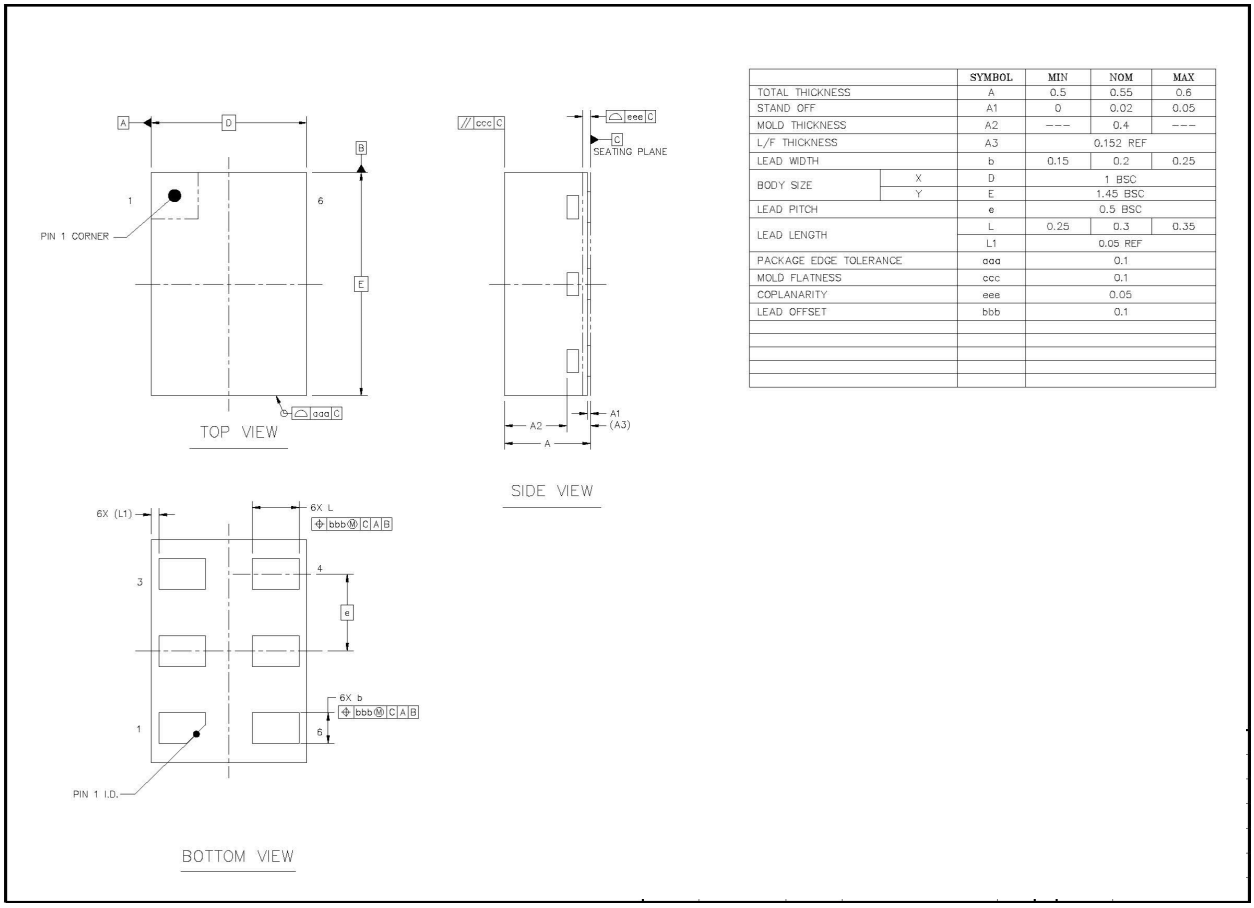
Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	0.900	1.100	0.035	0.043
A1	0.000	0.100	0.000	0.004
A2	0.900	1.000	0.035	0.039
b	0.150	0.350	0.006	0.014
c	0.110	0.175	0.004	0.007
D	2.000	2.200	0.079	0.087
E	1.150	1.350	0.045	0.053
E1	2.150	2.450	0.085	0.096
e	0.650 TYP.		0.026 TYP.	
e1	1.200	1.400	0.047	0.055
L	0.525 REF.		0.021 REF.	
L1	0.260	0.460	0.010	0.018
θ	0°	8°	0°	8°

DFN1x1-6L



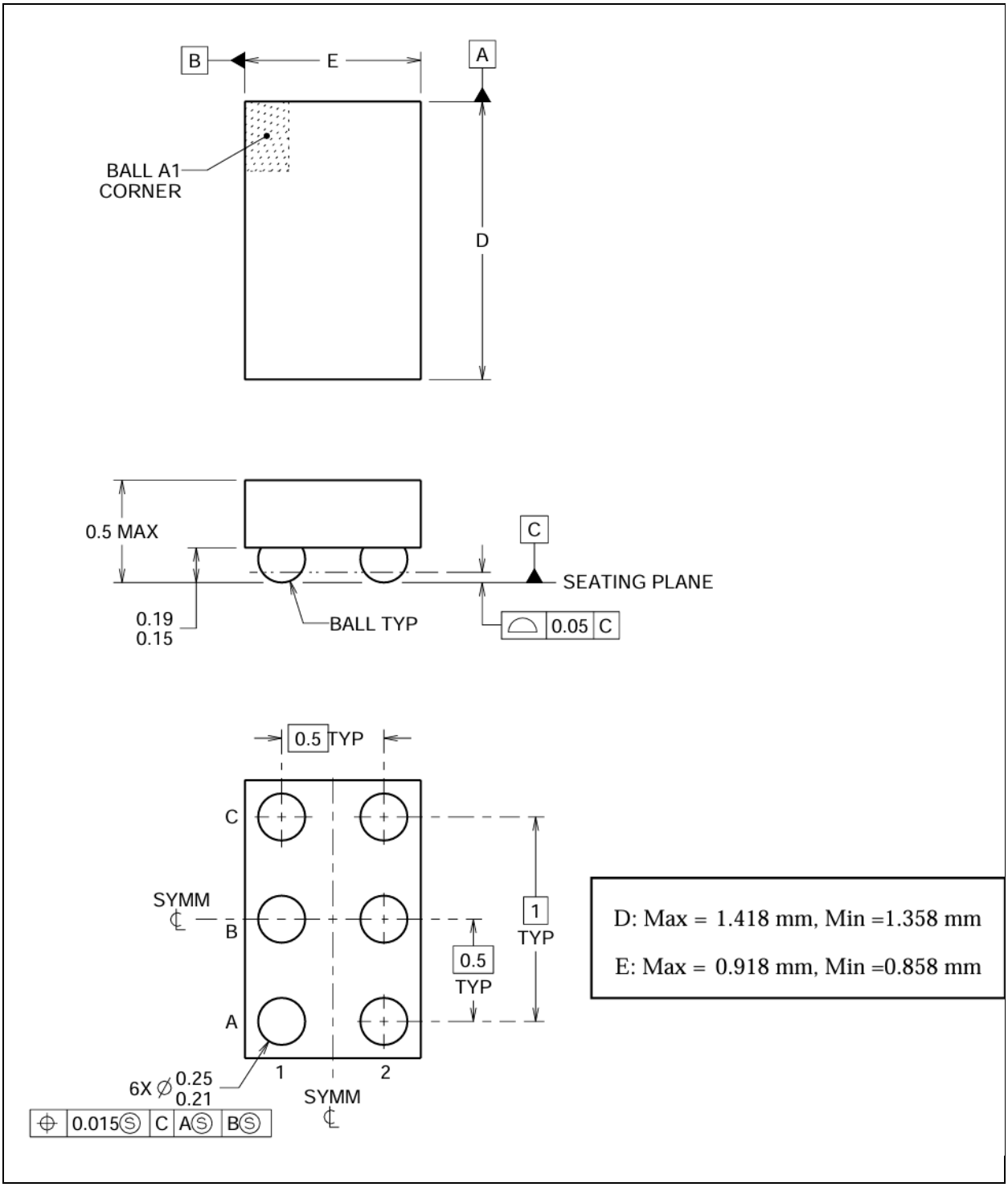
		SYMBOL	MIN	NOM	MAX
TOTAL THICKNESS		A	0.32	0.37	0.42
STAND OFF		A1	0	0.02	0.05
MOLD THICKNESS		A2	---	0.27	---
L/F THICKNESS		A3		0.102 REF	
BODY SIZE	X	D		1.00 BSC	
	Y	E		1.00 BSC	
LEAD WIDTH		b	0.10	0.15	0.20
LEAD PITCH		e		0.35 BSC	
		e1		0.55 BSC	
LEAD LENGTH		L	0.25	0.30	0.35
		L1	0.30	0.35	0.40

DFN1x1.45-6L



		SYMBOL	MIN	NOM	MAX
TOTAL THICKNESS		A	0.5	0.55	0.6
STAND OFF		A1	0	0.02	0.05
MOLD THICKNESS		A2	---	0.4	---
L/F THICKNESS		A3		0.152 REF	
LEAD WIDTH		b	0.15	0.2	0.25
BODY SIZE	X	D		1 BSC	
	Y	E		1.45 BSC	
LEAD PITCH		e		0.5 BSC	
LEAD LENGTH		L	0.25	0.3	0.35
		L1		0.05 REF	
PACKAGE EDGE TOLERANCE		aaa		0.1	
MOLD FLATNESS		ccc		0.1	
COPLANARITY		eee		0.05	
LEAD OFFSET		bbb		0.1	

BGA1.39x0.89-6B



12. Tape and Reel Information

12.1. Carrier tape dimensions

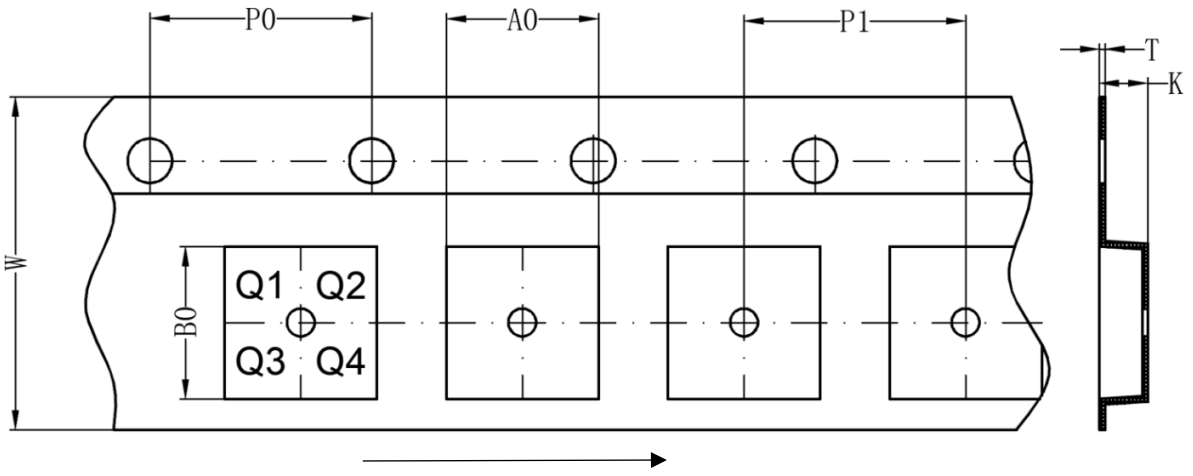


Table 12. Carrier tape dimensions

Package version	A0(mm)	B0(mm)	K(mm)	T(mm)	P1(mm)	W(mm)	P0(mm)	PIN 1
SOT23-6L	3.23	3.17	1.37	0.25	4	8	4	Q3
SOT363	2.55	2.55	1.2	0.2	4	8	4	Q3
DFN1x1-6L	1.16	1.16	0.5	0.23	4	8	4	Q1
DFN1x1.45-6L	1.15	1.6	0.75	0.2	4	8	4	Q1

12.2. Reel and box dimensions

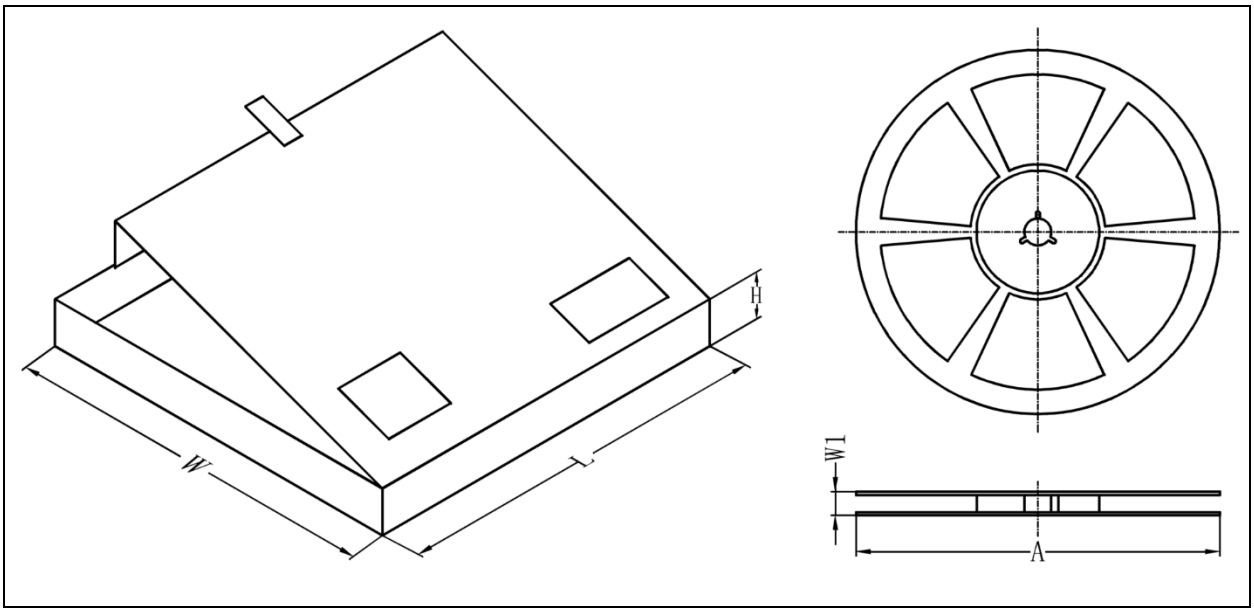


Table 13. Dimensions and quantities

Package version	Type NO. ending	MSL	Reel Dimension A (mm)	Max Reel Width W1 (mm)	SPQ (pcs)[1]	Reels per box	Outer box dimensions L×W×H(mm)[2]
SOT23-6L	GV	MSL1	180	13.2	3000	1	210x200x40
SOT363	GW	MSL1	180	13.2	3000	1	210x200x40
DFN1x1-6L	GS	MSL1	180	13.2	3000	1	210x200x40
DFN1x1.45-6L	GM	MSL1	180	13.2	3000	1	210x200x40

[1] Packing quantity dependent on specific product type. Please contact your local Energymath representative for ordering.

[2] Dimensions for reference only.

13. Abbreviations

Table 14. Abbreviations

Acronym	Description
CMOS	Complementary Metal-Oxide Semiconductor
DUT	Device Under Test
ESD	ElectroStatic Discharge
HBM	Human Body Model
CDM	Charged Device Model

14. Revision History

Table 15. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EM74LVC_LVCH1T45 Rev. 0.3	Oct 15.2026	Draft datashet		EM74LVC_LVCH1T45 Rev. 0.2
Modifications:	• Section 2 Features and Benefits updated.			
EM74LVC_LVCH1T45 Rev. 0.2	Jun 10, 2026	Draft datasheet		EM74LVC_LVCH1T45 Rev. 0.1
Modifications:	• Added BGA1.39X0.89-6B package.			
EM74LVC_LVCH1T45 Rev. 0.1	Feb 10, 2026	Draft datasheet		